

GHGT-12

The Peterhead-Goldeneye gas post-combustion CCS project

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Abstract

Full scale demonstration projects are the key to gaining policy acceptance for CCS. They also form the vital first step towards reducing the cost of clean power: only when a project is built can engineers examine it and identify improvements and savings opportunities. A number of projects are currently in development for coal CCS, but carbon capture on gas is also important if the world is to make the depth of cuts in emissions required by the current scientific consensus.

Shell and SSE plan to deliver the world's first full scale CCS project on gas in Scotland. Up to 10 million tonnes of carbon dioxide (CO₂) emissions could be captured over a 10-year period from flue-gas from a 400MW combined cycle gas turbine at the existing Peterhead Power Station in Aberdeenshire, Scotland and then transported by pipeline and stored, approximately 100km offshore in the depleted Goldeneye gas reservoir, at a depth of more than 2km under the floor of the North Sea. This would provide enough clean electricity to power the equivalent of 500,000 homes per year.

The project is one of two preferred bidders in the UK Government's CCS Commercialisation Programme and is currently in the midst of a front end engineering and design (FEED) phase. Subject to positive final investment decisions by Shell and the UK Government, and the receipt of all relevant permits and consents, the project is expected to be up and running by the end of the decade.

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1. Introduction

The Peterhead CCS project is the result of a partnership between Shell UK Limited, SSE Generation Limited, and the UK Government. Shell are the developers of the project, while SSE are strategic partners, as owners of the Peterhead Power Station in Aberdeenshire on which the project will be based.



Figure 1 : The location of the key elements of the Peterhead CCS project

In 2012 the UK Government published its CCS Roadmap. This set out a programme of measures and interventions to support industry in reducing the cost of CCS to a level where it will be competitive with other forms of low carbon generation. One area of this was the £1 billion CCS Commercialisation Programme. This programme included a Competition for FEED funding, a capital grant, sharing of CCS risks, and a negotiated contract for difference.

The competition was launched in April 2012 and there were eight bids. Two bids were selected to go forward to the FEED stage. These were the White Rose Project in Yorkshire, England: an oxy-fuel coal project with a saline aquifer store; and the Peterhead project. Both projects are expected to take Final Investment Decisions towards the end of 2015. Peterhead project

The Peterhead project is unique as it involves the capture of CO₂ from industrial scale gas-fired power. Natural gas is the cleanest-burning fossil fuel, and CCS can make it even cleaner, capturing and storing up to 90% of the CO₂ emissions generated from burning gas to create electricity. The project is a retrofit of an existing combined cycle gas turbine (CCGT) creating an important demonstration point of how to make existing generating stock even cleaner.

Retrofitting and reuse does not stop at the power station. The project will reuse an existing pipeline, the existing offshore platform, and wells. It will inject CO₂ into the now depleted Goldeneye gas field, which was a producing gas field until 2011. This makes the most of the UK's legacy oil and gas resources and provides a second proof point for reuse. The elements of the project are shown in Figure 1.

2. The Peterhead Project

2.1. Peterhead Power Station

The power station is located next to the village of Boddam in Aberdeenshire. It is almost at the easternmost tip of Great Britain. The station began operating in 1982 using oil. In 2000, the station completed a major repowering project to increase its efficiency and capacity. Three gas turbines were used in an innovative manner to provide steam to one of the original steam turbines. One of the existing gas turbines and associated heat recovery steam generators at the Peterhead Power Station is to be modified in order to provide a flue gas feedstock to the new Carbon Capture Plant. The steam generated from Heat Recovery Steam Generator (HRSG) will be used to drive a new steam turbine. The HRSG will be retrofitted with a new Selective Catalytic Unit in order to remove NO_x contaminants from the flue gas before it is sent to the CO_2 capture unit. Other modifications to the power station include the preparation of required utility connections, such as cooling water, demineralized water and town water, amongst others. Peterhead is one of the largest power stations in Scotland and is shown in Figure 2.



Figure 2: Peterhead Power Station

2.2. Carbon capture and compression

The project intends to capture the CO_2 from the output of one of the existing three gas turbines from downstream of the HRSG – effectively abating ~400 MWe (CCGT) of output (pre CCS retrofit). Figure 3 shows a block diagram of the process.

A booster fan is used to provide the flue gas with enough pressure to be fed to the Carbon Capture Plan via a direct contact cooler and gas/gas exchanger. A regenerable amine technology is used for the CO_2 capture plant and comprises a single very large absorber column, which will be made of concrete, and a smaller steel stripper column plus the associated pumps and heat exchangers. The capture unit will remove approximately 90% of the CO_2 present in the flue gas; equivalent to roughly 1 million tonnes per year. The capture plant uses steam from the new steam turbine, which will be installed at Peterhead Power Station, for amine regeneration.

The CO₂ product delivered from the capture plant to the compression and conditioning plant (see Figure 4), co-located at the Peterhead Power Station, will be water saturated and will contain traces of oxygen. The produced CO₂ stream will be cooled and partly compressed before having oxygen removed via catalytic reactions with hydrogen. Water is to be removed using molecular sieve technology. The conditioned CO₂ will then further be compressed to ~120 Bar(g), where it will be in dense phase and be ready for export to the Goldeneye platform.

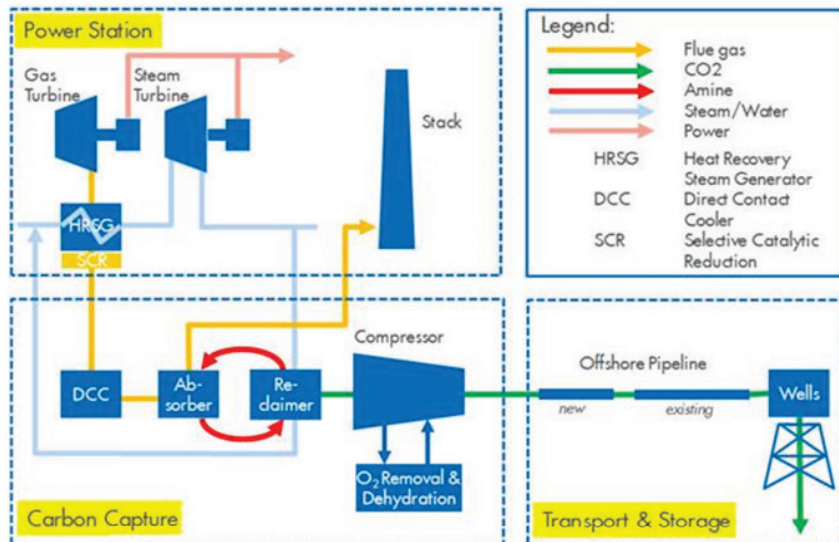


Figure 3: Block diagram of the full chain



Figure 4: Schematic of Peterhead with the capture plant in the background and the compression plant in the foreground.

Optimizations to minimize electrical and thermal energy used for the CO₂ capture and compression processes have been investigated and incorporated in the design, in an effort to maximize the net amount of power export by the power plant.

2.3. Transport

The current Goldeneye pipeline was installed in 2003 along with the platform. It is a 20-inch carbon steel line running from the Goldeneye platform to the St Fergus gas plant.

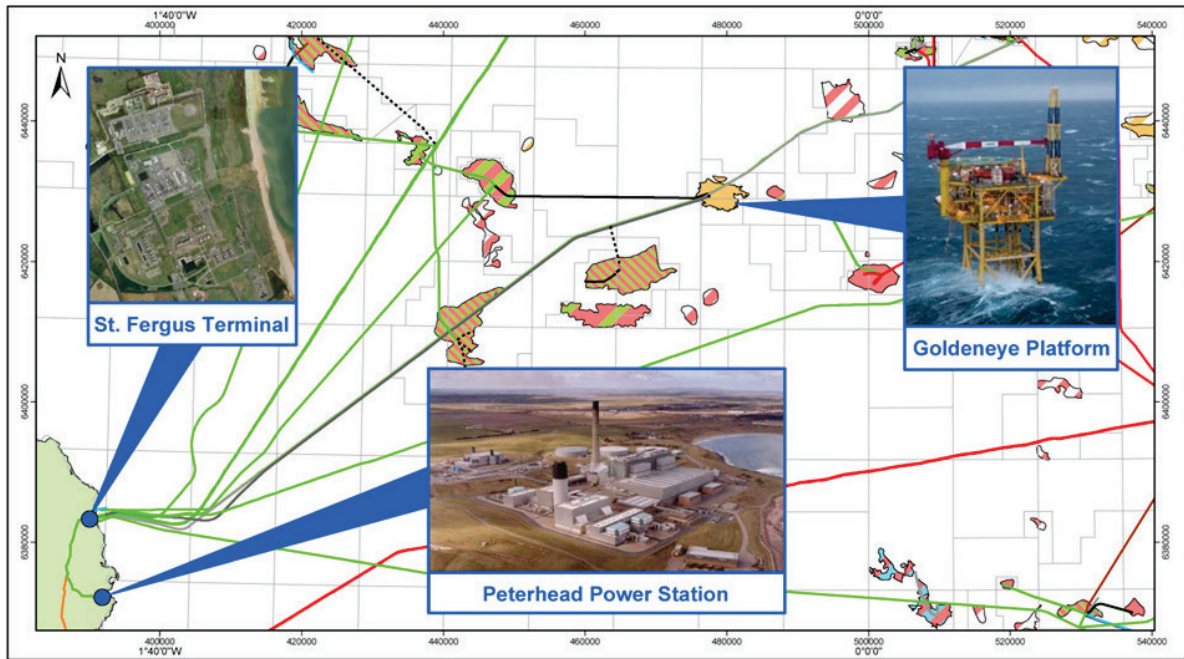


Figure 5 Existing infrastructure

Figure 5 shows the relationship of the St Fergus terminal to the platform and Peterhead Power Station. A new pipeline spur will be constructed from Peterhead to the existing Goldeneye pipeline. The diameter of the pipeline will be retained at 20 inches – the size of the existing gas export line. This means that significant overcapacity will exist in relation to the current duty. The current line includes a non return valve near the platform. This will be replaced with a subsea isolation valve as part of the refit.

2.4. The platform

The Goldeneye platform (Figure 6) consists of a four-legged steel structure, connected to the seabed with two vertical steel piles at each corner, that supports a topsides deck structure with a helideck, pedestal crane and vent stack. The jacket and topsides were installed in the field during 2003. It sits in 119m of water and is located 100km off the coast of Scotland. It is a “normally unmanned” installation and will remain so during the CCS project.

The platform needs to be upgraded to allow it to receive CO₂ - replacing piping, installing filters and advanced monitoring and control systems.



Figure 6: Goldeneye platform

2.5. Well engineering

There are five production wells and four decommissioned exploration and appraisal wells in the field. Four of the five production wells will be re-fitted for CO₂ service. This will require removal of the production tubing and production packer, and replacement with dual purpose design.

From an integrity perspective, the completion has to be able to take cold CO₂. When transported the CO₂ cools to ambient seabed temperature, around 6°C, while the produced gas was at around 80°C. Thermal contraction is therefore important, whereas before thermal expansion was key in the design.

The second purpose of the design is to provide back pressure to the CO₂ to maintain it in the dense phase. This, again, reduces cooling during injection.

The decommissioned wells have all been assessed and found to have excellent abandonment plugs at the reservoir level. They are therefore suited to the new CO₂ storage duty.

2.6. The storage reservoir

The CO₂ will be injected into the storage site at a depth >2516m [8255ft] below sea level into the previously gas-bearing portion of the high quality Captain Sandstone Member – in total a 130km long and <10km wide ribbon of Lower Cretaceous turbiditic sandstone fringing the southern margin of the South Halibut Shelf, from UKCS block 13/23 to block 21/2. At the Goldeneye field, this sandstone has permeability of between 700 and 1500mD. A schematic cross section is shown in Figure 7.

From 2004 to 2011, the field produced 568Bscf of gas and 23MMbbl of condensate. During production, the field experienced moderate to strong aquifer support – which also served to end the gas production from the wells as each well sequentially cut water.

The primary CO₂ storage mechanism will be accommodation in the pore space previously occupied by the produced gas and condensate from the Goldeneye field. A secondary mechanism will be immobile capillary

trapping in the water-leg below the original hydrocarbon accumulation. When CO₂ is injected into the field it will displace the invaded aquifer back into the aquifer. The CO₂ injection is shown in Figure 8.

A significant benefit of injecting into a depleted field is the ability to characterise the dynamic behaviour of the storage formation.

The UK regulator facilitated sharing of the pressure and production data from all fields in the hydraulically connected unit. This allowed the project to combine its own second-by-second bottom hole pressure data, with that from fields over 110km of the turbidite system, demonstrating lateral connectivity and the lack of compartmentalisation.

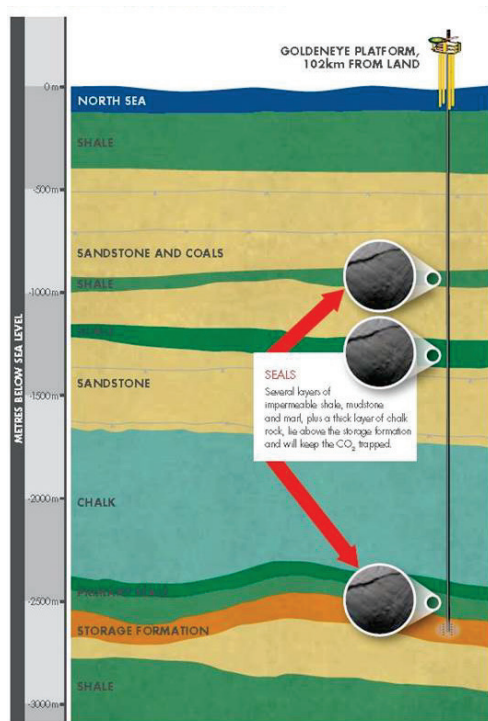


Figure 7: Storage and sealing reservoir units

3. Public consultation

Projects are not just about steel and concrete. They impact on the daily lives of the people who live near them and work on them. Shell has therefore been engaging, from a very early stage of the project, with the communities closest to the power station, in particular the people of Boddam and Peterhead. Two phases of public consultation have already taken place, in order to get feedback from the community on the project plans as they progress and to integrate this feedback into the planning and design work. A third phase of consultation will take place towards the end of 2015, prior to the submission of the onshore and offshore environmental impacts assessments.

Shell is eager to work closely with the local communities in order to bring benefits to the area as a result of this world first project and also to help create the right conditions to support the development of a wider CCS industry in the region into the future.

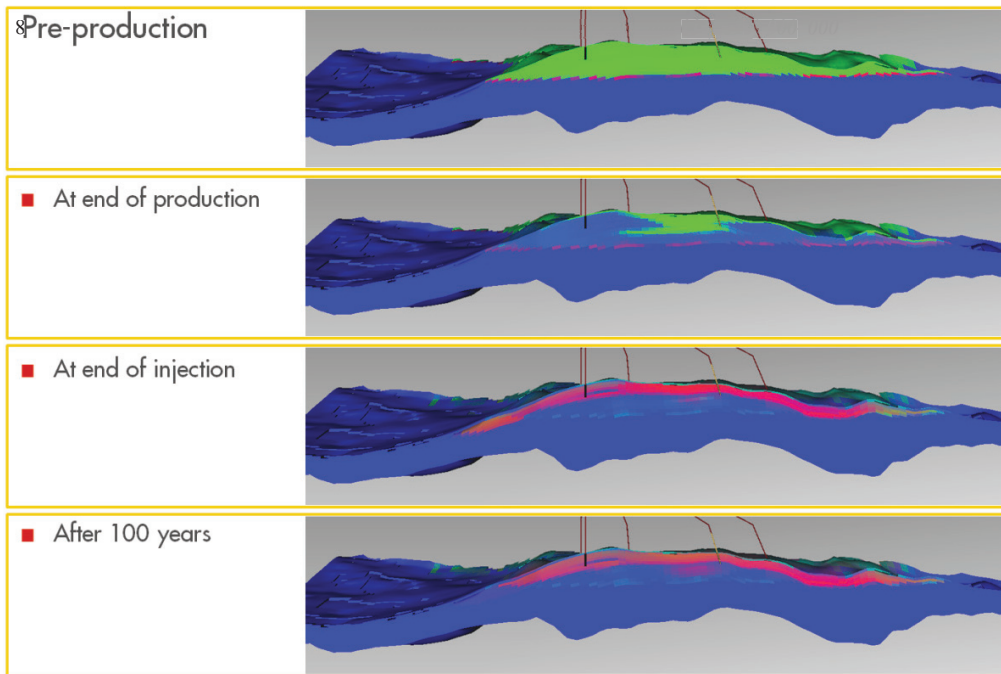


Figure 8: Hydrocarbon (green) production and CO₂ (red) injection.



Figure 9: Exploring CO₂ at the local school

Shell is also working with other key CCS partners – including Scottish Carbon Capture and Storage (SCCS), the Global CCS Institute and the Carbon Capture and Storage Association (CCSA) – and industry organizations to create broader awareness, both in Scotland and the UK, among the general public of what CCS is and the vital role it has to play in helping to address the climate change challenge in the decades to come.

4. Current status

The project has completed its subsurface characterisation and risk assessment work. It has explored the whole gamut of the geological, hydraulic, geochemical, geomechanical and thermal effects of CO₂ injection, and examined the effects of this on the well stock. It is preparing for the storage permit application.

From a facilities perspective, the project is currently in front end engineering design – for the power station modifications, the capture plant, the pipeline and the offshore modifications.

The key steps and anticipated timeline for the project are summarized in Figure 10. The hope is to start injection before the end of the decade.

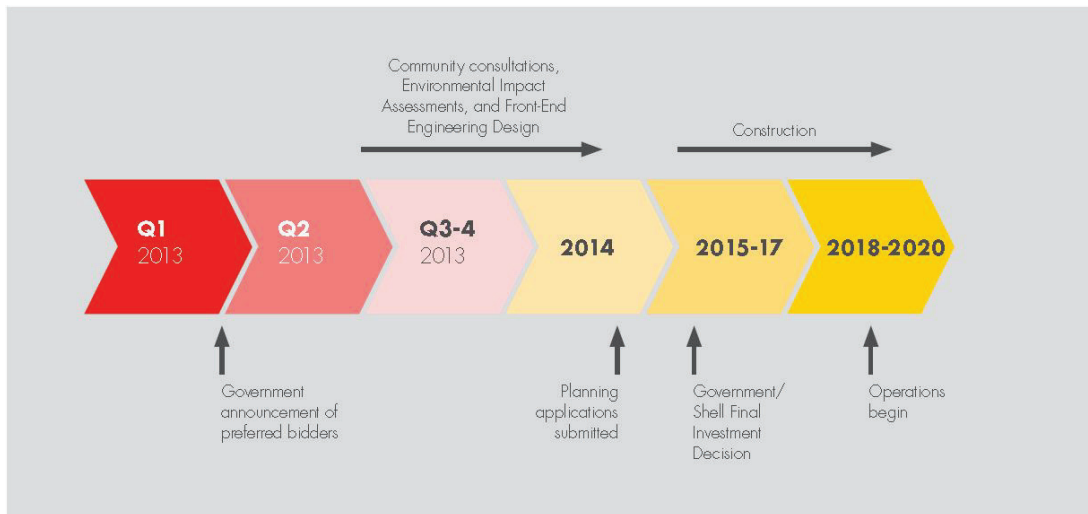


Figure 10: Project time line

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